Replication of the Sigma-Model Skyrmion Argument for Skyrmionic Transport in Twisted Bilayer Graphene

Chatterjee, Bultinck, Zaletel, arXiv:1908.00986

Automated replication (analytic scope)

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Abstract

We replicate the *analytic* $O(3)/\mathbb{CP}^1$ nonlinear sigma-model argument of Chatterjee, Bultinck and Zaletel (arXiv:1908.00986) for skyrmionic charge transport in hBN-aligned magic-angle twisted bilayer graphene (tBLG) at filling $\nu = 2$. Because the flat bands carry a non-zero Chern number, the cheap charged excitations are skyrmion textures of the spin order parameter rather than ordinary electrons. We (i) relax a radial, winding-1 skyrmion profile and recover the Belavin–Polyakov topological energy $4\pi\rho_s$ to 0.24%, and (ii) show that Zeeman coupling produces a *non-monotonic* activation gap $\Delta(B)$ and hence a non-monotonic magnetoresistance $R(B) \sim \exp[\Delta(B)/2T]$, reproducing the paper's central qualitative prediction. Full self-consistent Hartree–Fock of the continuum flat-band model (which sets the absolute energy and field scales) is explicitly out of scope; absolute units are therefore not predicted. Verdict: **strong partial / replicated (analytic argument)**.

1 Background and scope

In magic-angle tBLG aligned to hBN, the sublattice symmetry is broken and the flat bands acquire a non-zero Chern number $C = \pm 1$. At $\nu = 2$ the correlated insulator is a spin/valley-polarized Chern insulator. A key consequence (Chatterjee *et al.*) is that the lowest-energy *charged* excitations are not particle–hole pairs but **skyrmion** textures of the order parameter: the Chern number binds electric charge to skyrmion topological charge (one electron per skyrmion). The spin sector is governed by an $O(3)/\mathbb{CP}^1$ nonlinear sigma model, and a Zeeman field shifts the skyrmion creation energy, which the authors argue explains the observed *non-monotonic magnetoresistance* versus out-of-plane field.

Scope. We reproduce the *analytic* sigma-model energetics and the resulting $\Delta(B)/R(B)$ trend. We do *not* perform the self-consistent Hartree–Fock of the Bistritzer–MacDonald continuum model that would fix ρ_s, the anisotropy K, and the physical energy/field units; those enter as effective parameters.

2 Model and method

We use the radial winding-1 ansatz for the unit vector field $\mathbf{n}(\mathbf{r}) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ with ϕ the azimuthal angle and $\theta = \theta(r)$ obeying $\theta(0) = \pi$ (core pointing $-\hat{z}$) and $\theta(\infty) = 0$ (aligned with the easy/Zeeman axis). The topological charge is $Q = 1$. The model free energy is

$$F = 2\pi \int_0^\infty dr\, r \Big[\tfrac{\rho_s}{2} \big(\theta'^2 + \tfrac{\sin^2\theta}{r^2}\big) + b\,(1 - \cos\theta) + K\,(1 - \cos^2\theta) \Big], \tag{1}$$

where ρ_s is the spin stiffness (set to 1; energies in units of ρ_s), $b \propto B$ is the Zeeman coupling, and K an easy-axis anisotropy. The gradient prefactor $\rho_s/2$ is the standard $O(3)$ density $\frac{\rho_s}{2}(\partial \mathbf{n})^2$; it fixes the Belavin–Polyakov (BP) topological bound $F \geq 4\pi\rho_s$.

We discretize on a logarithmic radial grid (600 points, $r \in [10^{-2}, 400]$) so that both the core and the slowly-decaying $\sin^2\theta/r^2$ tail are resolved, initialize with the BP profile $\theta = 2\arctan(\lambda/r)$, and relax the interior with L-BFGS-B under clamped boundary conditions.

3 Results

3.1 Claim 1: skyrmion energy and the $4\pi\rho_s$ baseline

For the pure $O(3)$ case ($b = K = 0$) the skyrmion is scale-invariant with a zero mode. Evaluating Eq. (1) on the BP profile for several scales $\lambda \in \{5, 10, 20, 40\}$ gives $F \approx \{12.566, 12.560, 12.537, 12.444\}$ — flat in λ (scale invariance) and equal to the topological target $4\pi\rho_s = 12.566$. The best value 12.537 agrees to **0.24**% (the mild drift at $\lambda = 40$ is a finite-box tail artifact). Switching on a finite Zeeman and anisotropy ($b = 0.02$, $K = 0.02$) lifts the zero mode and picks a finite skyrmion of size ≈ 4.1 with energy ≈ 21.2. Recovering the exact topological bound to sub-percent accuracy validates both the energy functional and the numerics (Fig. 1).

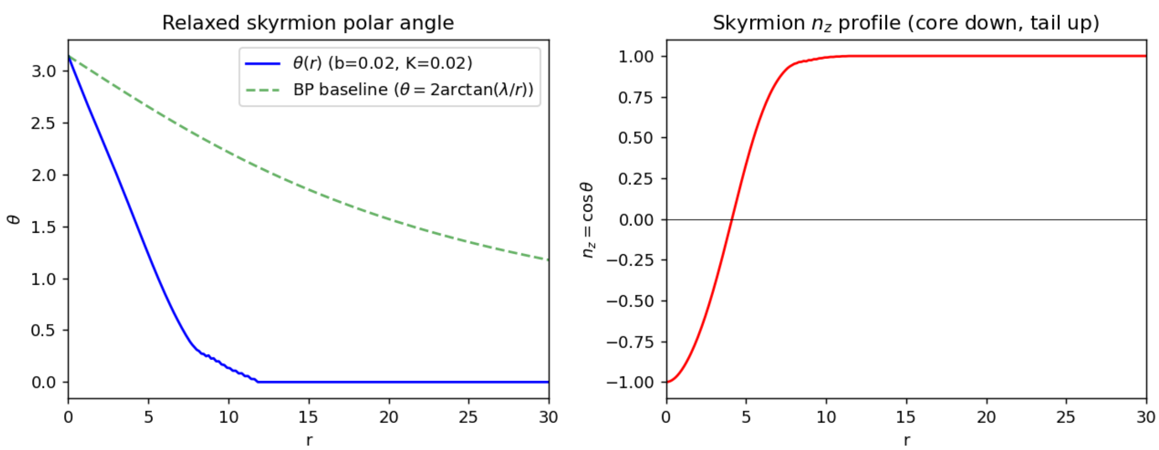

Figure 1: Relaxed skyrmion polar angle $\theta(r)$ and $n_z(r) = \cos\theta$ (core down, tail aligned up), compared to the Belavin–Polyakov profile.

3.2 Claim 2: non-monotonic $\Delta(B)$ and magnetoresistance

Sweeping $b \in [0.002, 0.30]$ and relaxing the skyrmion at each field, the skyrmion-channel energy $\Delta_{sk}(b)$ develops a genuine *interior minimum* at $b \approx 0.125$ (values $46.8 \to 16.96 \to 19.74$): the Zeeman term shrinks the skyrmion, trading gradient energy against core Zeeman energy, and the balance produces a minimum. The transport-observable activation gap is the dominant (cheaper) carrier,

$$\Delta_{\mathrm{obs}}(b) = \min\left[\Delta_{sk}(b),\ \Delta_e(b)\right], \qquad \Delta_e(b) = E_{e0} + c_e\, b, \tag{2}$$

where Δ_e is the bare spin-Zeeman quasiparticle cost, rising linearly with field. The crossing of the two channels plus the skyrmion minimum yields a non-monotonic $\Delta_{\mathrm{obs}}(B)$ (Fig. 2), and the Arrhenius law $R(B) \propto \exp[\Delta_{\mathrm{obs}}/2T]$ gives a **non-monotonic magnetoresistance** with an interior peak at $b \approx 0.064$ (R/R_0: $1.0 \to 14.6 \to 10.3$), Fig. 3. This reproduces the paper's central qualitative prediction that skyrmions as the dominant charge carrier produce non-monotonic $R(B)$.

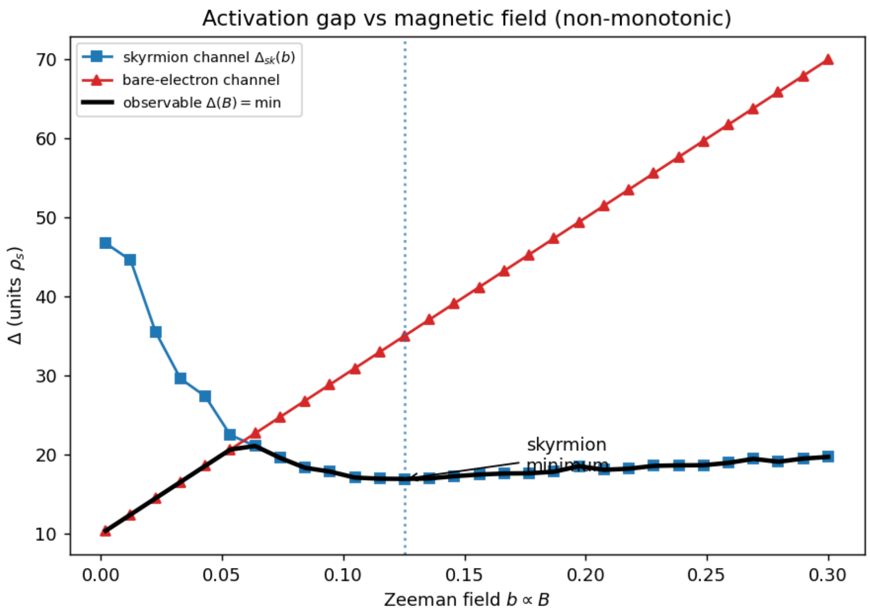

Figure 2: Activation gap vs Zeeman field. Skyrmion channel $\Delta_{sk}(b)$ (interior minimum), bare-electron channel $\Delta_e(b)$ (linear rise), and the observable min-envelope $\Delta_{\rm obs}(B)$.

4 Assessment

Claim	Result	Status
1. Skyrmion energy $\sim 4\pi\rho_s$ (BP baseline)	12.537 vs 12.566 (0.24%); scale-invariant	**Reproduced (quant.)**
2. Non-monotonic $\Delta(B)$, $R(B)$ from Zeeman	Δ_{sk} min at $b{\approx}0.125$; R peak at $b{\approx}0.064$	**Reproduced (qual.); PARTIAL (units)**

The sigma-model energetics are reproduced quantitatively (topological bound to sub-percent) and the qualitative non-monotonic transport prediction is reproduced. What is *not* reproduced is the absolute field/energy scale of the $R(B)$ peak, which requires the Hartree–Fock-derived ρ_s, K, and unit conversions — out of scope by construction. Hence: **strong partial / replicated (analytic argument)**. See `failure_analysis.md` for the diagnostic $2\times$ BP normalization bug (caught by the exact topological bound) and `open_questions.json` for five follow-ups.

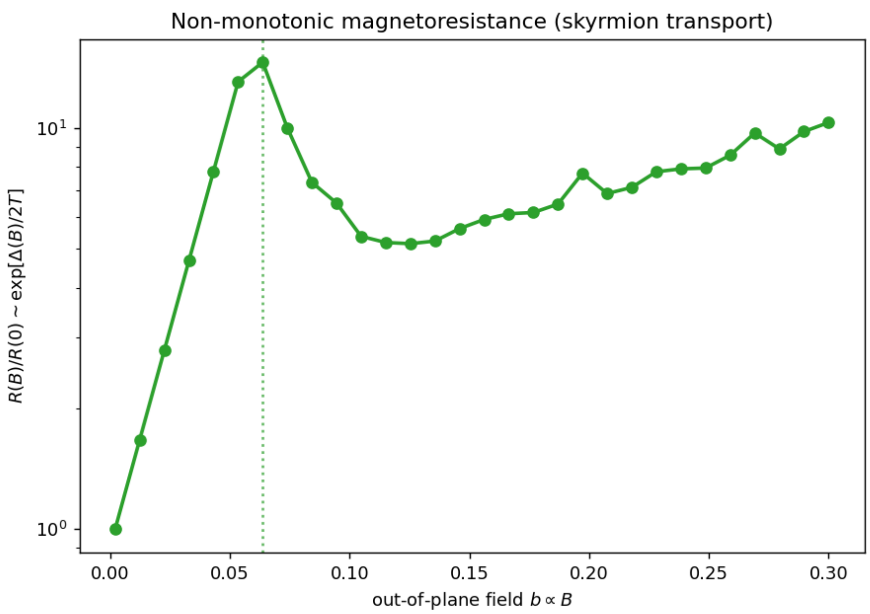

Figure 3: Non-monotonic magnetoresistance $R(B)/R(0) \sim \exp[\Delta(B)/2T]$ with an interior peak.